CARBON DATA LOGGER

Personal Carbon Footprint Tracking System

Project Handbook and Session Record

CBSE Class XII Computer Science (083) · Python + MySQL

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Repository: github.com/SG1222/Stuff · Updated 12 October 2026

1. Project brief

The project started from an agent brief (prompt.pdf) and the CBSE Class XII Computer Science syllabus for 2026–27 (CS_Board_Syllabus.pdf). The brief's main points were:

Objective

Build a terminal-based Python + MySQL program that lets one person record everyday activities that add to their carbon footprint (electricity, car/bus/train travel, LPG, flights). Python calculates the estimated CO2 with predefined emission factors and stores the records in MySQL. The user can search, edit, delete, summarise and export the data. The purpose is educational, so the emission factors must be labelled as illustrative.

Requirements

Area	What the brief asked for
Level	Within CBSE Class XI/XII concepts. No classes/OOP, decorators, generators, threading, APIs, web frameworks, ORMs, Pandas/NumPy, complex comprehensions or design patterns.
Size	About 500–650 meaningful lines across the Python and SQL files, without padding.
Interface	A neat terminal menu (13 options + Exit). No GUI, no login.
Database	Database carbon_logger with two related tables: categories (name, unit, emission factor) and emission_records (date, category, quantity, CO2, note), linked by a foreign key.
Calculation	CO2 = quantity × emission factor, in one function; factors read from MySQL; rounded to 2 decimals.
Features	Add (with confirmation), view (via JOIN), search (ID, category, date, date range), update (recalculate CO2), delete (with Y/N), overall/category/monthly summaries, highest category, rule-based suggestions, CSV export, text summary export, manage factors, help.
SQL to show	CREATE DATABASE/TABLE, PRIMARY/FOREIGN KEY, INSERT, SELECT, UPDATE, DELETE, WHERE, BETWEEN, ORDER BY, COUNT/SUM/AVG/MIN/MAX, GROUP BY, INNER JOIN.
Connectivity	mysql.connector with connect(), cursor(), execute(), commit(), fetchone(), fetchall(); parameterised %s queries.
Errors	try/except/finally for bad menu input, non-numeric/negative quantity, invalid dates, missing IDs, MySQL connection and file errors.
Files	carbon_logger.py (whole app) and database_setup.sql; the program generates carbon_records.csv and carbon_summary.txt.
Test	Could a Class XII student explain every part in a viva? If not, simplify.

2. The finished project at a glance

Repository files

File	Purpose
README.md (repo root)	Full explanation of the project, written in Simplified Technical English, including 22 'tricky parts' sections.
CarbonDataLogger/carbon_logger.py	The program: 45 functions, about 635 non-blank lines.
CarbonDataLogger/database_setup.sql	Creates the database, both tables, 6 default categories and 5 sample records.
CarbonDataLogger/VIVA_SHEET.md	61 likely viva questions with answers, plus a demo order.
CarbonDataLogger/Project_Report.docx	The CBSE project report (39 pages): certificate, design, code, 18 screenshots, 25 tests, bibliography.
CarbonDataLogger/Carbon_Data_Logger_Handbook.pdf	This handbook: brief, run guide, session record, tests and full code.
.gitignore	Keeps generated files (CSV, text summary, __pycache__) out of the repo.

Main menu

Option	Feature	Option	Feature
1	Add Carbon Record	8	Monthly Summary
2	View All Records	9	Highest Emission Category
3	Search Records (sub-menu)	10	Export Records to CSV
4	Update Record	11	Export Summary to Text File
5	Delete Record	12	Manage/View Emission Factors (sub-menu)
6	Carbon Footprint Summary	13	Help
7	Category-wise Summary	14	Undo Last Added Record (stack)
		0	Exit

Database design

Table	Columns	Keys
categories	category_id INT, category_name VARCHAR(30), unit VARCHAR(10), emission_factor FLOAT	PRIMARY KEY category_id (AUTO_INCREMENT); category_name UNIQUE
emission_records	record_id INT, record_date DATE, category_id INT, quantity FLOAT, co2_emission FLOAT, note VARCHAR(100)	PRIMARY KEY record_id (AUTO_INCREMENT); FOREIGN KEY category_id → categories

Default categories (illustrative factors, kg CO2 per unit): Electricity 0.82/kWh · Car Travel 0.17/km · Bus Travel 0.10/km · Train Travel 0.04/km · LPG 2.98/kg · Flight 0.15/km.

How the code is organised

- Constants at the top: MySQL settings (`DB_PASSWORD = "root"`), file names, default categories, suggestion texts, month names, and the shared RECORD_QUERY (SELECT … INNER JOIN).
- Setup: `connect_database()`, `create_tables()`, `insert_default_categories()`.
- Input helpers that repeat until the input is valid: `get_integer()`, `get_positive_number()`, `validate_date()`, `get_date()`.

- One function per feature (`add_record()`, `search_by_date_range()`, `monthly_summary()` …) and shared helpers such as `print_records()` and `get_category_totals()`.
- Undo uses a list as a stack with `push()`/`pop()`; `add_record()` pushes `cursor.lastrowid`.
- `main()` connects, then runs the menu loop through the dictionary `menu_actions`; database errors return to the menu; Ctrl+C and Exit close the connection in `finally`.

3. How to run the project on your own device

You need three things: Python 3, a MySQL server, and the Python package mysql-connector-python. The steps below are for Windows; macOS/Linux differences are at the end.

Step 1 – Install Python 3

1. Download Python 3 from python.org and run the installer.
2. On the first screen, tick **“Add python.exe to PATH”**, then click Install.
3. Check it: open Command Prompt and type `python --version`.

Step 2 – Install MySQL Server

1. Download **MySQL Installer (Community)** from dev.mysql.com and choose MySQL Server 8 (MySQL Workbench is optional).
2. During setup, set a **root password** and write it down. Using **root** means you don't need to change the code.
3. Leave “Start MySQL Server at System Startup” ticked so the server is running.

Step 3 – Get the project files

Either download the ZIP from GitHub (repository page → green **Code** button → **Download ZIP**, then extract it), or clone it:

```
git clone https://github.com/SG1222/Stuff.git
```

The program files are in the folder `CarbonDataLogger`.

Step 4 – Install the MySQL connector

```
pip install mysql-connector-python
```

If `pip` is not found, use `python -m pip install mysql-connector-python`.

Step 5 – Set your MySQL password

Open `carbon_logger.py` in IDLE or VS Code and find the line `DB_PASSWORD = "root"`. If your MySQL root password is not **root**, type your password between the quotes and save.

Step 6 (optional) – Load the sample data

This gives you 5 sample records to demonstrate straight away. Open Command Prompt in the CarbonDataLogger folder (in File Explorer, type **cmd** in the address bar and press Enter), then run:

```
mysql -u root -p < database_setup.sql
```

Enter your MySQL password when asked. If `mysql` is not recognised, open **MySQL Command Line Client** from the Start menu and type the SOURCE command with the full path, using forward slashes, for example:

```
SOURCE C:/Users/YourName/Downloads/Stuff/CarbonDataLogger/database_setup.sql;
```

Run the script only once. A second run stops with “Duplicate entry”. To start again, run `DROP DATABASE carbon_logger;` in the MySQL client and run the script again. If you skip this step, the program creates an empty database with the 6 categories by itself.

Both the script and the program create the database with `CHARACTER SET utf8mb4`, so notes can contain ₹ and Hindi text. If you created `carbon_logger` with an older version of this project, drop it

first so it is made again with utf8mb4.

Step 7 – Run the program

In the CarbonDataLogger folder:

```
python carbon_logger.py
```

Type a menu number and press Enter. Choose 0 to exit. Exported files (carbon_records.csv, carbon_summary.txt) are created in the folder you ran the program from.

macOS / Linux

- Install Python 3 and MySQL (macOS: from python.org and dev.mysql.com, or with Homebrew: `brew install python mysql`, then `brew services start mysql`).
- Use `python3` and `pip3` instead of `python` and `pip`.
- Load sample data with `mysql -u root -p < database_setup.sql` in Terminal.

Troubleshooting

What you see	Cause	Fix
ModuleNotFoundError: No module named 'mysql'	Connector not installed	pip install mysql-connector-python
Could not connect to MySQL … Access denied	Wrong password	Change DB_PASSWORD in carbon_logger.py
Could not connect to MySQL … Can't connect	MySQL server not running	Start it (Windows: Services → MySQL80 → Start)
'python' is not recognized	Python not on PATH	Use py carbon_logger.py, or reinstall with “Add to PATH” ticked
Duplicate entry … when running the SQL script	Data already loaded	DROP DATABASE carbon_logger; then run the script again
Database error … Incorrect string value (note with ₹ or Hindi)	Old latin1 database	DROP DATABASE carbon_logger; then start again (README 10.22)
Invalid date! for a date that looks right	Date before 2000 or after today	Type a date from 2000 up to today, or press Enter for today
Error: could not write to carbon_records.csv	File open in Excel	Close the file, then export again
The < redirect fails in PowerShell	PowerShell syntax	Use Command Prompt (cmd), or the SOURCE command

4. Session record

Each of your messages is shown in a blue box, followed by a summary of what was done. This is a structured record of the conversation, not a word-for-word transcript of the assistant's replies.

Step 1

You: Shared prompt.pdf (agent brief) and CS_Board_Syllabus.pdf. “The two files contain everything needed for the task. Proceed.”

Done: Read both PDFs. Built carbon_logger.py and database_setup.sql to the brief (two related tables, JOIN, aggregates, 13-option menu, exports, error handling). Installed MariaDB and the connector in the cloud environment and tested every menu option with scripted input, including bad inputs. Committed and pushed to branch claude/wizardly-hamilton-6q6g0h.

Step 2

You: use float, set password to root

Done: Changed the numeric columns from DECIMAL to FLOAT in both files. The password was already root. Re-tested and pushed.

Step 3

You: Review code once again, check for hidden bugs and clean any unwanted lines. Before edits please confirm.

Done: Reviewed line by line and tested suspected cases. Reported 4 bugs (a long note lost the record; nan/inf accepted; export crashed on ₹/Hindi text on Windows; Ctrl+C showed a traceback) and 4 cleanups (duplicate grand-total loop, an extra query, validate_date called twice, .title() damaging names). Made no edits, as asked.

Step 4

You: which all are critical?

Done: Ranked them: #3 (export crash) critical; #1 moderate; #2 low; #4 cosmetic; 5–8 not bugs.

Step 5

You: 1,3,4,5 to 8 apply (make sure edits are within CBSE scope)

Done: Applied them with syllabus-level code only: [:100] slicing and length checks, encoding="utf-8", except KeyboardInterrupt, a get_grand_total() helper, and the small tidy-ups. Tested each and pushed. #2 was left as chosen.

Step 6

You: Is the code ship and presentation ready?

Done: Yes, with a pre-demo checklist: check DB_PASSWORD, install the connector, load the SQL script once, and do a practice run on the actual PC. Noted that CBSE also expects a project report.

Step 7

You: Asked to add the suggested viva sheet to the repo.

Done: Wrote VIVA_SHEET.md (46 questions at the time) and a demo order, then pushed.

Step 8

You: Shared simplify.skill (ASD-STE100 Simplified Technical English): “update the readme with the entire project explained especially the tricky bits.”

Done: Wrote README.md in STE style: glossary, setup with Warning/Caution notes, menu table, database design, program flow, 18 'tricky parts' sections, SQL map, troubleshooting and limitations.

Step 9

> **You:** merge with main

Done: There was no main branch yet, so main was created from the work branch. Noted that GitHub's default branch must be switched to main in Settings.

Step 10

> **You:** Asked for the full CBSE report (introduction, table structures, code, screenshots, bibliography) as one document in the repo.

Done: Captured real program output, rendered terminal-style screenshots and built Project_Report.docx: cover, certificate, acknowledgement, index, 12 sections, full code, figures, tests and bibliography. Fixed a broken LibreOffice install to check the pages. Pushed to both branches.

Step 11

> **You:** also add a PDF version of the report

Done: Added Project_Report.pdf (fonts embedded).

Step 12

> **You:** nvm, remove the pdf keep docx

Done: Removed the PDF and its README line.

Step 13

> **You:** Screenshot of GitHub's empty “Add a README” box: “the readme file should come here instead”

Done: Moved README.md to the repository root, made the file list clickable, and said which folder to open in the setup steps.

Step 14

> **You:** merge with main

Done: Checked: main and the work branch were already identical.

Step 15

> **You:** what is gitignore file for

Done: Explained that it lists files Git should not track: the generated CSV and text summary and Python's __pycache__ folder.

Step 16

> **You:** Shared another version (carbon_logger.py, emission_factors.csv, requirements.txt, setup.sql): “compare yours with this. No edits only comparison. Which is better suited”

Done: Compared them against the brief and the syllabus. Ours is better suited: the other has one table (no foreign key, JOIN or GROUP BY), no search, update or text export, fixed dates, matplotlib, and advanced Python (type hints, pathlib, defaultdict, lambda). Its best idea was an undo feature using a stack (a syllabus topic).

Step 17

> **You:** i think stack option is good, use it

Done: Added menu option 14, Undo Last Added Record: undo_stack list, push() and pop() functions, LIFO undo with confirmation, plus the empty-stack and already-deleted cases. Updated the README, viva sheet (new Stack section) and report (new screenshot, functions and tests). Merged a README edit you had made on GitHub instead of overwriting it.

Step 18

You: is the code still within CBSE scope. After answering this run a last test run and report.

Done: Scope: only csv, datetime and mysql.connector; no classes, lambdas, comprehensions, f-strings or type hints. Wrote an automated end-to-end test: 45/45 checks passed (the 4 early failures were mistakes in the test script). pyflakes clean. Synced your second GitHub README edit.

Step 19

You: wait, what is lastrowid? you said its out of syllabus

Done: Clarified: cursor.lastrowid gives the AUTO_INCREMENT ID of the row just inserted. It isn't on the syllabus list, but it is the same kind of attribute as rowcount. Offered SELECT MAX(record_id) as a syllabus-only alternative.

Step 20

You: keep lastrowid no changes, does the read me and viva sheet contain all the details?

Done: Audited both files against the code and found 8 gaps (connector calls, x if c else y, NULL aggregates, fetchall()[0], MONTH_NAMES[month-1], lastrowid and the syllabus, the nan limitation).

Step 21

You: add all

Done: Added README sections 10.20 and 10.21 and five viva questions, updated the limitations, and renumbered the viva sheet to 59 questions. Pushed.

Step 22

You: How does one run the project on own device? Export all our chat and everything into a pdf … (don't push to repo) Include a brief of the project and code to provide background info

Done: This handbook (first version), kept out of the repository at the time.

Step 23

You: merge with main

Done: Checked: main and the work branch were already identical (the handbook was not in the repo).

Step 24

You: check for bugs

Done: Re-ran all 45 checks under MySQL 8's stricter default settings (all passed) and probed new edge cases. Found: (A) ₹/Hindi notes rejected on servers that default to latin1, such as MySQL 5.7; (B) Ctrl+D / Ctrl+Z showed an EOFError traceback; (C) future dates accepted; (D) a year typed as 0026 silently became 2026; (E) quantities like 1e39 rejected by MySQL; (F) the zero-value message said 'Quantity' for emission factors. Made no edits.

Step 25

You: ACDBF update all documentation as well. then merge with main

Done: Fixed A (CREATE DATABASE … CHARACTER SET utf8mb4), B (except (KeyboardInterrupt, EOFError)), C+D (validate_date accepts 2000 up to today) and F (neutral message). Added 10 tests: 55/55 passed under both MariaDB and MySQL 8 modes. Updated the README (new section 10.22), viva sheet (61 questions) and report (39 pages, 25 tests, new screenshots). Merged into main. E was left as before.

Step 26

> **You:** yes, upload in repo asw

Done: Updated this handbook to the current version and added it to the repository.

Your own changes on GitHub

- Edited the README introduction to “It is my CBSE Class XII Computer Science project…” and removed the STE note.
- Removed the README limitation line about CSV number formatting (10.0, 1.7).
- Added a disclaimer at the end of the README: “Please note, this is just a small project, data produced may be inaccurate.”
- All three edits were merged into the work branch and kept.

5. Key decisions, test results and known limits

Design decisions

Decision	Reason
Two tables + foreign key + INNER JOIN	Required by the brief; shows relational concepts from the syllabus.
Emission factors stored in MySQL	Changeable from menu 12 without editing code.
Old records keep their CO2 when a factor changes	A record shows the estimate made at the time.
FLOAT columns	Your choice; values display cleanly with {:.2f}.
%s placeholders everywhere	Prevents SQL injection and handles quotes in text.
cursor.lastrowid kept	Standard and short; same kind as rowcount (documented in README 10.20 and the viva sheet).
Undo with a list stack (push/pop)	Covers the syllabus Stack topic in a real feature.
Notes cut to 100 characters	The column is VARCHAR(100); long notes no longer lose the record.
UTF-8 for exports	₹ and Hindi text no longer crash the export on Windows.
Database created with CHARACTER SET utf8mb4	Older servers default to latin1, which rejects ₹ and Hindi text in notes.
Dates only from 2000 up to today	Rejects future dates and typing mistakes such as 0026-09-15.
Ctrl+D / Ctrl+Z handled like Ctrl+C	Ends the program cleanly and still closes the connection.

Final test run (55 of 55 checks passed, under both MariaDB and MySQL 8 default settings)

Area	Checked
Startup	Program creates the DB and categories itself; the SQL script loads 6 categories + 5 records
Empty database	View, summaries 6–9, exports and undo show proper messages
Add	Bad category, impossible date, text/negative/zero quantity, blank date = today, cancel, 150-char note, ₹/Hindi note
Dates	Future, pre-2000 and 0026 dates rejected; today and 2000-01-01 accepted; update keeps the old date for a future one
View / Search	JOIN output; by ID, category (LIKE), date, date range with swap (BETWEEN); invalid sub-menu choice

Area	Checked
Update / Delete	Missing ID, invalid date kept, CO2 recalculated, bad quantities rejected; delete cancel/confirm
Summaries	Overall, category share %, monthly (invalid month, empty month), highest + suggestions
Exports	CSV header + all rows with Unicode intact; text summary complete
Factors	View, update, add, duplicate, too long, empty; new factor applies to new records only
Undo (stack)	LIFO order, cancel, already-deleted record, empty stack
Character set	Program and SQL script both create the database as utf8mb4; ₹/Hindi note saved
Other	Help, invalid main-menu input, wrong password, Ctrl+C, Ctrl+D/Ctrl+Z (end of input), real 'python carbon_logger.py' run; pyflakes clean

Known limits

- Emission factors are illustrative, not official data.
- Single user, no login, terminal only.
- Undo works only for records added since the program started (the stack is kept in memory).
- Typing nan, inf or a huge number such as 1e39 as a quantity shows a database error; the program does not crash.
- A carbon_logger database made with an older version on a latin1 server must be dropped once to get utf8mb4.
- Size is about 686 lines (635 Python + 51 SQL), slightly over the brief's “approximately 500–650”.

Before you submit

1. Fill in the [bracketed] placeholders in Project_Report.docx (school, name, section, roll number, teacher, principal).
2. Check DB_PASSWORD on the presentation PC.
3. Optional: make main the default branch in GitHub → Settings → General.

6. Source code

6.1 database_setup.sql

59 lines

```
 1  -- ============================================================
 2  -- Carbon Data Logger - Database Setup Script
 3  -- Creates the database, both tables and the default categories.
 4  -- Run ONCE in the MySQL client:   SOURCE database_setup.sql;
 5  -- (carbon_logger.py also creates these automatically if missing.)
 6  -- ============================================================
 7
 8  -- utf8mb4 lets notes contain symbols such as the rupee sign and Hindi text
 9  CREATE DATABASE IF NOT EXISTS carbon_logger CHARACTER SET utf8mb4;
10  USE carbon_logger;
11
12  -- Table 1: activity categories with their unit and emission factor
13  CREATE TABLE IF NOT EXISTS categories (
14      category_id     INT AUTO_INCREMENT PRIMARY KEY,
15      category_name   VARCHAR(30) NOT NULL UNIQUE,
16      unit            VARCHAR(10) NOT NULL,
17      emission_factor FLOAT NOT NULL
18  );
19
20  -- Table 2: one row for every activity recorded by the user
21  CREATE TABLE IF NOT EXISTS emission_records (
22      record_id    INT AUTO_INCREMENT PRIMARY KEY,
23      record_date  DATE NOT NULL,
```

```
    category_id  INT NOT NULL,
    quantity     FLOAT NOT NULL,
    co2_emission FLOAT NOT NULL,
    note         VARCHAR(100),
    FOREIGN KEY (category_id) REFERENCES categories(category_id)
);

-- Default categories.
-- NOTE: emission factors (kg CO2 per unit) are ILLUSTRATIVE project
-- values for learning purposes only. They can be changed from menu 12.
INSERT INTO categories (category_name, unit, emission_factor) VALUES
    ('Electricity',  'kWh', 0.820),
    ('Car Travel',   'km',  0.170),
    ('Bus Travel',   'km',  0.100),
    ('Train Travel', 'km',  0.040),
    ('LPG',          'kg',  2.980),
    ('Flight',       'km',  0.150);

-- Some sample records for testing (optional)
INSERT INTO emission_records (record_date, category_id, quantity, co2_emission, note) VALUES
    ('2026-08-02', 1, 12.50, 10.25, 'Daily home use'),
    ('2026-08-03', 2, 20.00,  3.40, 'Trip to market'),
    ('2026-08-05', 3, 15.00,  1.50, 'School bus'),
    ('2026-09-01', 5,  2.00,  5.96, 'Cooking gas'),
    ('2026-09-10', 6, 800.00, 120.00, 'Flight to Mumbai');

-- Useful queries for checking the data
SELECT * FROM categories;

SELECT r.record_id, r.record_date, c.category_name, r.quantity, c.unit, r.co2_emission
FROM emission_records r INNER JOIN categories c ON r.category_id = c.category_id
ORDER BY r.record_date;

SELECT c.category_name, COUNT(*), SUM(r.co2_emission)
FROM emission_records r INNER JOIN categories c ON r.category_id = c.category_id
GROUP BY c.category_name;
```

6.2 carbon_logger.py

763 lines

```
# ============================================================
# CARBON DATA LOGGER - Personal Carbon Footprint Tracking System
# CBSE Class XII Computer Science Project (Python + MySQL)
#
# Records everyday activities (electricity, travel, LPG, flights),
# estimates their CO2 emissions and stores them in MySQL.
# Emission factors are ILLUSTRATIVE project values, not official data.
# ============================================================

import csv
import datetime
import mysql.connector

# MySQL connection settings - change the password to match your system
DB_HOST = "localhost"
DB_USER = "root"
DB_PASSWORD = "root"
DB_NAME = "carbon_logger"

# Output file names
CSV_FILE = "carbon_records.csv"
SUMMARY_FILE = "carbon_summary.txt"

# Default categories: (name, unit, illustrative kg CO2 per unit)
DEFAULT_CATEGORIES = [
    ("Electricity", "kWh", 0.82),
    ("Car Travel", "km", 0.17),
    ("Bus Travel", "km", 0.10),
    ("Train Travel", "km", 0.04),
    ("LPG", "kg", 2.98),
    ("Flight", "km", 0.15),
]

# Simple rule-based suggestions for each category
SUGGESTIONS = {
    "Electricity": "Switch off unused lights/fans and prefer LED bulbs and efficient appliances.",
    "Car Travel": "Consider public transport, car-pooling or cycling for suitable journeys.",
    "Bus Travel": "Bus travel is already a good choice; walk or cycle for very short trips.",
    "Train Travel": "Train travel is a low-emission option; keep using it where possible.",
    "LPG": "Use lids while cooking and pressure cookers to save cooking gas.",
    "Flight": "Flights add a lot of CO2; prefer trains for shorter distances when possible.",
}

MONTH_NAMES = ("January", "February", "March", "April", "May", "June", "July",
               "August", "September", "October", "November", "December")

# Common SELECT used to show records with category name and unit (JOIN)
RECORD_QUERY = ("SELECT r.record_id, r.record_date, c.category_name, r.quantity, "
                "c.unit, r.co2_emission, r.note "
                "FROM emission_records r INNER JOIN categories c "
                "ON r.category_id = c.category_id ")

connection = None
cursor = None

# Stack of record IDs added in this session, used by "Undo Last Added Record"
undo_stack = []


# ---------------- Database setup ----------------

# Connect to the MySQL server and select (or create) the database
def connect_database():
    global connection, cursor
    connection = mysql.connector.connect(host=DB_HOST, user=DB_USER, password=DB_PASSWORD)
    cursor = connection.cursor()
    # utf8mb4 lets notes contain symbols such as the rupee sign and Hindi text
    cursor.execute("CREATE DATABASE IF NOT EXISTS " + DB_NAME + " CHARACTER SET utf8mb4")
    cursor.execute("USE " + DB_NAME)


# Create both tables if they do not exist yet
def create_tables():
    cursor.execute("""CREATE TABLE IF NOT EXISTS categories (
                          category_id INT AUTO_INCREMENT PRIMARY KEY,
                          category_name VARCHAR(30) NOT NULL UNIQUE,
```

```
                        unit VARCHAR(10) NOT NULL,
                        emission_factor FLOAT NOT NULL)""")
    cursor.execute("""CREATE TABLE IF NOT EXISTS emission_records (
                        record_id INT AUTO_INCREMENT PRIMARY KEY,
                        record_date DATE NOT NULL,
                        category_id INT NOT NULL,
                        quantity FLOAT NOT NULL,
                        co2_emission FLOAT NOT NULL,
                        note VARCHAR(100),
                        FOREIGN KEY (category_id) REFERENCES categories(category_id))""")
    connection.commit()


# Insert the default categories only when the table is empty
def insert_default_categories():
    cursor.execute("SELECT COUNT(*) FROM categories")
    count = cursor.fetchone()[0]
    if count == 0:
        query = "INSERT INTO categories (category_name, unit, emission_factor) VALUES (%s, %s, %s)"
        for category in DEFAULT_CATEGORIES:
            cursor.execute(query, category)
        connection.commit()
        print("Default categories added.")


# ---------------- Input helper functions ----------------

# Wait for the user before showing the menu again
def pause_program():
    input("\nPress Enter to continue...")


def print_heading(title):
    print("\n" + "=" * 60)
    print(title.center(60))
    print("=" * 60)


# Keep asking until the user types a whole number
def get_integer(prompt):
    while True:
        try:
            return int(input(prompt))
        except ValueError:
            print("Invalid input! Please enter a whole number.")


# Keep asking until the user types a number greater than zero
def get_positive_number(prompt):
    while True:
        try:
            value = float(input(prompt))
            if value <= 0:
                print("The value must be greater than zero.")
            else:
                return value
        except ValueError:
            print("Invalid input! Please enter a numeric value.")


# Return the date as 'YYYY-MM-DD' if it is valid, otherwise None.
# Only dates from the year 2000 up to today are accepted.
def validate_date(date_text):
    try:
        date_value = datetime.datetime.strptime(date_text, "%Y-%m-%d").date()
    except ValueError:
        return None
    if date_value.year < 2000 or date_value > datetime.date.today():
        return None
    return str(date_value)


# Ask for a date; an empty answer means today's date
def get_date(prompt):
    while True:
        date_text = input(prompt + " (YYYY-MM-DD, Enter for today): ").strip()
        if date_text == "":
            return str(datetime.date.today())
        valid_date = validate_date(date_text)
        if valid_date is None:
```

```
157             print("Invalid date! Use YYYY-MM-DD, from the year 2000 up to today.")
158         else:
159             return valid_date
160
161
162 def ask_yes_no(prompt):
163     answer = input(prompt + " (Y/N): ").strip().upper()
164     return answer == "Y"
165
166
167 # ---------------- Categories ----------------
168
169 # Display all categories in a table and return them as a list
170 def display_categories():
171     cursor.execute("SELECT category_id, category_name, unit, emission_factor "
172                    "FROM categories ORDER BY category_id")
173     categories = cursor.fetchall()
174     print("\n{:<5}{:<16}{:<8}{:>22}".format("ID", "CATEGORY", "UNIT", "FACTOR (kg CO2/unit)"))
175     print("-" * 51)
176     for category in categories:
177         print("{:<5}{:<16}{:<8}{:>22.3f}".format(category[0], category[1],
178                                                 category[2], category[3]))
179     return categories
180
181
182 # Fetch one category (id, name, unit, factor) from MySQL, or None
183 def get_category(category_id):
184     cursor.execute("SELECT category_id, category_name, unit, emission_factor "
185                    "FROM categories WHERE category_id = %s", (category_id,))
186     return cursor.fetchone()
187
188
189 # Ask the user to pick a valid category ID
190 def choose_category():
191     display_categories()
192     while True:
193         category_id = get_integer("\nEnter category ID: ")
194         category = get_category(category_id)
195         if category is None:
196             print("No category with that ID. Try again.")
197         else:
198             return category
199
200
201 # ---------------- Carbon calculation ----------------
202
203 # Calculate carbon emission: CO2 = quantity x emission factor
204 def calculate_emission(quantity, emission_factor):
205     return round(float(quantity) * float(emission_factor), 2)
206
207
208 # ---------------- Add and view records ----------------
209
210 def add_record():
211     print_heading("ADD CARBON RECORD")
212     category = choose_category()
213     record_date = get_date("Enter date")
214     quantity = get_positive_number("Enter quantity in " + category[2] + ": ")
215     # Only the first 100 characters are kept because the note column is VARCHAR(100)
216     note = input("Enter a note (optional): ").strip()[:100]
217
218     # The emission factor comes from the categories table in MySQL
219     co2 = calculate_emission(quantity, category[3])
220     print("\nCalculation:", quantity, category[2], "x", category[3], "kg/" + category[2],
221           "=", co2, "kg CO2")
222
223     if ask_yes_no("Save this record?"):
224         query = ("INSERT INTO emission_records (record_date, category_id, quantity, "
225                  "co2_emission, note) VALUES (%s, %s, %s, %s, %s)")
226         values = (record_date, category[0], quantity, co2, note)
227         cursor.execute(query, values)
228         connection.commit()
229         print("Record saved successfully with ID", cursor.lastrowid)
230         # Remember the new record so that it can be undone later
231         push(undo_stack, cursor.lastrowid)
232     else:
233         print("Record not saved.")
234
235
236 # Print a list of records as a neat table
```

```
237 def print_records(records):
238     if len(records) == 0:
239         print("\nNo records found.")
240         return
241     print("\n{:<5}{:<13}{:<15}{:>10}  {:<6}{:>10}  {}".format(
242         "ID", "DATE", "CATEGORY", "QUANTITY", "UNIT", "CO2 (kg)", "NOTE"))
243     print("-" * 80)
244     for record in records:
245         note = record[6] if record[6] else ""
246         print("{:<5}{:<13}{:<15}{:>10.2f}  {:<6}{:>10.2f}  {}".format(
247             record[0], str(record[1]), record[2], record[3], record[4], record[5], note))
248     print("-" * 80)
249     print("Records shown:", len(records))
250 
251 
252 # Display all stored records
253 def view_all_records():
254     print_heading("ALL CARBON RECORDS")
255     cursor.execute(RECORD_QUERY + "ORDER BY r.record_date, r.record_id")
256     records = cursor.fetchall()
257     print_records(records)
258 
259 
260 # Fetch one record by its ID, or None if it does not exist
261 def get_record(record_id):
262     cursor.execute(RECORD_QUERY + "WHERE r.record_id = %s", (record_id,))
263     return cursor.fetchone()
264 
265 
266 # ---------------- Search ----------------
267 
268 def search_by_id():
269     record_id = get_integer("Enter record ID: ")
270     record = get_record(record_id)
271     if record is None:
272         print("No record found with ID", record_id)
273     else:
274         print_records([record])
275 
276 
277 def search_by_category():
278     name = input("Enter category name (or part of it): ").strip()
279     cursor.execute(RECORD_QUERY + "WHERE c.category_name LIKE %s ORDER BY r.record_date",
280                    ("%" + name + "%",))
281     print_records(cursor.fetchall())
282 
283 
284 def search_by_date():
285     record_date = get_date("Enter date")
286     cursor.execute(RECORD_QUERY + "WHERE r.record_date = %s", (record_date,))
287     print_records(cursor.fetchall())
288 
289 
290 def search_by_date_range():
291     start_date = get_date("Enter start date")
292     end_date = get_date("Enter end date")
293     if start_date > end_date:
294         print("Start date was after end date, so the dates have been swapped.")
295         start_date, end_date = end_date, start_date
296     cursor.execute(RECORD_QUERY + "WHERE r.record_date BETWEEN %s AND %s "
297                    "ORDER BY r.record_date", (start_date, end_date))
298     print_records(cursor.fetchall())
299 
300 
301 def search_menu():
302     while True:
303         print_heading("SEARCH RECORDS")
304         print("1. Search by Record ID")
305         print("2. Search by Category")
306         print("3. Search by Date")
307         print("4. Search between two dates")
308         print("5. Return to Main Menu")
309         choice = input("\nEnter your choice: ").strip()
310         if choice == "1":
311             search_by_id()
312         elif choice == "2":
313             search_by_category()
314         elif choice == "3":
315             search_by_date()
316         elif choice == "4":
```

```
            search_by_date_range()
        elif choice == "5":
            return
        else:
            print("Invalid choice! Please enter 1 to 5.")
        pause_program()


# ---------------- Update and delete ----------------

def update_record():
    print_heading("UPDATE RECORD")
    record_id = get_integer("Enter record ID to update: ")
    record = get_record(record_id)
    if record is None:
        print("No record found with ID", record_id)
        return
    print("\nCurrent details:")
    print_records([record])

    # The category ID is not part of RECORD_QUERY, so read it separately
    cursor.execute("SELECT category_id FROM emission_records WHERE record_id = %s",
                   (record_id,))
    category_id = cursor.fetchone()[0]
    quantity = record[3]
    new_date = str(record[1])
    new_note = record[6]

    print("\nPress Enter to keep the current value.")
    date_text = input("New date (YYYY-MM-DD): ").strip()
    if date_text != "":
        valid_date = validate_date(date_text)
        if valid_date is None:
            print("Invalid date, keeping the old date.")
        else:
            new_date = valid_date

    if ask_yes_no("Change the category?"):
        category_id = choose_category()[0]

    quantity_text = input("New quantity: ").strip()
    if quantity_text != "":
        try:
            if float(quantity_text) > 0:
                quantity = float(quantity_text)
            else:
                print("Quantity must be positive, keeping the old quantity.")
        except ValueError:
            print("Not a number, keeping the old quantity.")

    note_text = input("New note: ").strip()[:100]
    if note_text != "":
        new_note = note_text

    # Recalculate CO2 using the factor of the (possibly new) category
    category = get_category(category_id)
    new_co2 = calculate_emission(quantity, category[3])

    query = ("UPDATE emission_records SET record_date = %s, category_id = %s, "
             "quantity = %s, co2_emission = %s, note = %s WHERE record_id = %s")
    cursor.execute(query, (new_date, category_id, quantity, new_co2, new_note, record_id))
    connection.commit()
    print("\nRecord updated. Recalculated CO2:", new_co2, "kg")
    print_records([get_record(record_id)])


def delete_record():
    print_heading("DELETE RECORD")
    record_id = get_integer("Enter record ID to delete: ")
    record = get_record(record_id)
    if record is None:
        print("No record found with ID", record_id)
        return
    print_records([record])
    if ask_yes_no("Delete this record?"):
        cursor.execute("DELETE FROM emission_records WHERE record_id = %s", (record_id,))
        connection.commit()
        print(cursor.rowcount, "record deleted.")
    else:
        print("Delete cancelled.")
```

```
397
398
399 # ---------------- Undo using a stack ----------------
400
401 # Add an item to the top of the stack
402 def push(stack, item):
403     stack.append(item)
404
405
406 # Remove and return the top item of the stack, or None if it is empty
407 def pop(stack):
408     if len(stack) == 0:
409         return None
410     return stack.pop()
411
412
413 # Remove the most recently added record (Last In, First Out)
414 def undo_last_record():
415     print_heading("UNDO LAST ADDED RECORD")
416     record_id = pop(undo_stack)
417     if record_id is None:
418         print("Nothing to undo. No records have been added in this session.")
419         return
420     record = get_record(record_id)
421     if record is None:
422         print("Record", record_id, "was already deleted.")
423         return
424     print_records([record])
425     if ask_yes_no("Remove this record?"):
426         cursor.execute("DELETE FROM emission_records WHERE record_id = %s", (record_id,))
427         connection.commit()
428         print("Record removed.", len(undo_stack), "more record(s) can be undone.")
429     else:
430         # Put the ID back so the user can undo it later
431         push(undo_stack, record_id)
432         print("Undo cancelled.")
433
434
435 # ---------------- Summaries ----------------
436
437 # Return (count, total, average, minimum, maximum) using aggregate functions
438 def get_overall_totals():
439     cursor.execute("SELECT COUNT(*), SUM(co2_emission), AVG(co2_emission), "
440                    "MIN(co2_emission), MAX(co2_emission) FROM emission_records")
441     return cursor.fetchone()
442
443
444 # Return a list of (category, number of records, total CO2), highest first
445 def get_category_totals():
446     cursor.execute("SELECT c.category_name, COUNT(*), SUM(r.co2_emission) "
447                    "FROM emission_records r INNER JOIN categories c "
448                    "ON r.category_id = c.category_id "
449                    "GROUP BY c.category_name ORDER BY SUM(r.co2_emission) DESC")
450     return cursor.fetchall()
451
452
453 # Add up the total CO2 of all categories
454 def get_grand_total(category_totals):
455     grand_total = 0
456     for row in category_totals:
457         grand_total = grand_total + float(row[2])
458     return grand_total
459
460
461 def overall_summary():
462     print_heading("CARBON FOOTPRINT SUMMARY")
463     count, total, average, lowest, highest = get_overall_totals()
464     if count == 0:
465         print("No records available yet.")
466         return
467     print("Total records              :", count)
468     print("Total CO2 emissions        : {:.2f} kg".format(total))
469     print("Average CO2 per record     : {:.2f} kg".format(average))
470     print("Lowest individual emission : {:.2f} kg".format(lowest))
471     print("Highest individual emission: {:.2f} kg".format(highest))
472
473     cursor.execute("SELECT MIN(record_date), MAX(record_date) FROM emission_records")
474     first_date, last_date = cursor.fetchone()
475     print("Records from", first_date, "to", last_date)
476
```

```
477
478 def category_summary():
479     print_heading("CATEGORY-WISE SUMMARY")
480     rows = get_category_totals()
481     if len(rows) == 0:
482         print("No records available yet.")
483         return
484     grand_total = get_grand_total(rows)
485     print("{:<18}{:>10}{:>16}{:>12}".format("CATEGORY", "RECORDS", "TOTAL CO2", "SHARE"))
486     print("-" * 56)
487     for row in rows:
488         share = float(row[2]) / grand_total * 100 if grand_total > 0 else 0
489         print("{:<18}{:>10}{:>16.2f}{:>11.1f}%".format(row[0], row[1], row[2], share))
490     print("-" * 56)
491     print("{:<18}{:>10}{:>16.2f}".format("TOTAL", "", grand_total))
492
493
494 # Show which months have records, using DISTINCT
495 def show_available_months():
496     cursor.execute("SELECT DISTINCT YEAR(record_date), MONTH(record_date) "
497                    "FROM emission_records ORDER BY 1, 2")
498     months = cursor.fetchall()
499     if len(months) > 0:
500         print("Months with records:")
501         for year, month in months:
502             print("  ", MONTH_NAMES[month - 1], year, "(" + str(month) + "/" + str(year) + ")")
503     return len(months)
504
505
506 def monthly_summary():
507     print_heading("MONTHLY SUMMARY")
508     if show_available_months() == 0:
509         print("No records available yet.")
510         return
511     month = get_integer("\nEnter month number (1-12): ")
512     if month < 1 or month > 12:
513         print("Invalid month! It must be between 1 and 12.")
514         return
515     year = get_integer("Enter year (e.g. 2026): ")
516
517     cursor.execute("SELECT COUNT(*), SUM(co2_emission), AVG(co2_emission) FROM emission_records "
518                    "WHERE MONTH(record_date) = %s AND YEAR(record_date) = %s", (month, year))
519     count, total, average = cursor.fetchone()
520     print("\nCARBON SUMMARY -", MONTH_NAMES[month - 1].upper(), year)
521     if count == 0:
522         print("No records for this month.")
523         return
524     print("\nTotal records:", count)
525     print("Total estimated CO2: {:.2f} kg".format(total))
526     print("Average emission: {:.2f} kg".format(average))
527
528     # Category with the largest total in that month
529     cursor.execute("SELECT c.category_name, SUM(r.co2_emission) "
530                    "FROM emission_records r INNER JOIN categories c "
531                    "ON r.category_id = c.category_id "
532                    "WHERE MONTH(r.record_date) = %s AND YEAR(r.record_date) = %s "
533                    "GROUP BY c.category_name ORDER BY SUM(r.co2_emission) DESC", (month, year))
534     top = cursor.fetchall()[0]
535     print("Highest category:", top[0], "({:.2f} kg)".format(top[1]))
536
537
538 # Return a list of suggestion strings based on the recorded categories
539 def generate_suggestions(category_totals):
540     suggestions = []
541     grand_total = get_grand_total(category_totals)
542     for row in category_totals:
543         name = row[0]
544         share = float(row[2]) / grand_total * 100 if grand_total > 0 else 0
545         # Suggest only for categories contributing at least 20% of the total
546         if share >= 20:
547             if name in SUGGESTIONS:
548                 advice = SUGGESTIONS[name]
549             else:
550                 advice = "Try to reduce this activity where it is practical."
551             suggestions.append(name + " ({:.1f}% of total): ".format(share) + advice)
552     return suggestions
553
554
555 def highest_emission_category():
556     print_heading("HIGHEST EMISSION CATEGORY")
```

```
    rows = get_category_totals()
    if len(rows) == 0:
        print("No records available yet.")
        return
    # Rows are ordered by total CO2 in descending order, so the first is the highest
    print("Category with the highest emissions:", rows[0][0])
    print("Total CO2 from this category: {:.2f} kg in {} record(s)".format(rows[0][2], rows[0][1]))
    if len(rows) > 1:
        print("Lowest category:", rows[-1][0], "({:.2f} kg)".format(rows[-1][2]))

    print("\nSuggestions:")
    for suggestion in generate_suggestions(rows):
        print("-", suggestion)


# ---------------- Export ----------------

# Write all records to a CSV file using the csv module
def export_to_csv():
    print_heading("EXPORT RECORDS TO CSV")
    cursor.execute(RECORD_QUERY + "ORDER BY r.record_date, r.record_id")
    records = cursor.fetchall()
    if len(records) == 0:
        print("No records to export.")
        return
    try:
        with open(CSV_FILE, "w", newline="", encoding="utf-8") as file:
            writer = csv.writer(file)
            writer.writerow(["Record ID", "Date", "Category", "Quantity", "Unit",
                             "CO2 (kg)", "Note"])
            writer.writerows(records)
        print(len(records), "records exported to", CSV_FILE)
    except IOError:
        print("Error: could not write to", CSV_FILE, "(is it open in another program?)")


# Write an overall and category-wise summary to a text file
def export_summary_to_text():
    print_heading("EXPORT SUMMARY TO TEXT FILE")
    count, total, average, lowest, highest = get_overall_totals()
    if count == 0:
        print("No records available to summarise.")
        return
    category_rows = get_category_totals()
    file = None
    try:
        file = open(SUMMARY_FILE, "w", encoding="utf-8")
        file.write("CARBON DATA LOGGER\n")
        file.write("PERSONAL FOOTPRINT SUMMARY\n")
        file.write("Generated on: " + str(datetime.date.today()) + "\n\n")
        file.write("Total Records: " + str(count) + "\n")
        file.write("Total Estimated CO2: {:.2f} kg\n".format(total))
        file.write("Average CO2: {:.2f} kg\n".format(average))
        file.write("Lowest Emission: {:.2f} kg\n".format(lowest))
        file.write("Highest Emission: {:.2f} kg\n\n".format(highest))
        file.write("CATEGORY SUMMARY\n")
        file.write("----------------\n")
        for row in category_rows:
            file.write("{}: {:.2f} kg ({} records)\n".format(row[0], row[2], row[1]))
        file.write("\nSUGGESTIONS\n")
        file.write("-----------\n")
        for suggestion in generate_suggestions(category_rows):
            file.write("- " + suggestion + "\n")
        file.write("\nNote: emission factors are illustrative project values.\n")
        print("Summary exported to", SUMMARY_FILE)
    except IOError:
        print("Error: could not write to", SUMMARY_FILE)
    finally:
        # The file is closed whether or not an error occurred
        if file is not None:
            file.close()


# ---------------- Manage emission factors ----------------

def update_emission_factor():
    category = choose_category()
    print("Current factor for", category[1], "is", category[3], "kg CO2 per", category[2])
    new_factor = get_positive_number("Enter new emission factor: ")
    cursor.execute("UPDATE categories SET emission_factor = %s WHERE category_id = %s",
```

```
                    (new_factor, category[0]))
    connection.commit()
    print("Emission factor updated. It will be used for new or edited records.")


def add_category():
    name = input("Enter new category name: ").strip()
    unit = input("Enter unit (e.g. kWh, km, kg): ").strip()
    if name == "" or unit == "":
        print("Name and unit cannot be empty.")
        return
    if len(name) > 30 or len(unit) > 10:
        print("Name can have at most 30 characters and unit at most 10.")
        return
    factor = get_positive_number("Enter emission factor (kg CO2 per " + unit + "): ")
    try:
        cursor.execute("INSERT INTO categories (category_name, unit, emission_factor) "
                       "VALUES (%s, %s, %s)", (name, unit, factor))
        connection.commit()
        print("Category", name, "added.")
    except mysql.connector.IntegrityError:
        print("A category with this name already exists.")


def manage_emission_factors():
    while True:
        print_heading("MANAGE / VIEW EMISSION FACTORS")
        print("Note: these factors are illustrative values for this project.")
        print("1. View Emission Factors")
        print("2. Update an Emission Factor")
        print("3. Add a New Category")
        print("4. Return to Main Menu")
        choice = input("\nEnter your choice: ").strip()
        if choice == "1":
            display_categories()
        elif choice == "2":
            update_emission_factor()
        elif choice == "3":
            add_category()
        elif choice == "4":
            return
        else:
            print("Invalid choice! Please enter 1 to 4.")
        pause_program()


# ---------------- Help and main menu ----------------

def show_help():
    print_heading("HELP")
    print("* Add a record for an activity, e.g. 15 kWh of electricity or 20 km by car.")
    print("* CO2 is estimated as: quantity x emission factor of the category.")
    print("* Dates are entered as YYYY-MM-DD, from 2000 up to today. Press Enter for today.")
    print("* Use Search, Update and Delete with the record ID shown in the tables.")
    print("* Summaries show totals overall, per category and per month.")
    print("* Exports create", CSV_FILE, "and", SUMMARY_FILE, "in this folder.")
    print("* Emission factors are illustrative and can be changed from option 12.")
    print("* Option 14 removes records added in this session, newest first.")


def display_main_menu():
    print("\n" + "=" * 50)
    print("CARBON DATA LOGGER".center(50))
    print("=" * 50)
    print("1. Add Carbon Record")
    print("2. View All Records")
    print("3. Search Records")
    print("4. Update Record")
    print("5. Delete Record")
    print("6. Carbon Footprint Summary")
    print("7. Category-wise Summary")
    print("8. Monthly Summary")
    print("9. Highest Emission Category")
    print("10. Export Records to CSV")
    print("11. Export Summary to Text File")
    print("12. Manage/View Emission Factors")
    print("13. Help")
    print("14. Undo Last Added Record")
    print("0. Exit")

```

```
717
718 def main():
719     try:
720         # Connect to the MySQL database and prepare the tables
721         connect_database()
722         create_tables()
723         insert_default_categories()
724     except mysql.connector.Error as error:
725         print("Could not connect to MySQL:", error)
726         print("Check that MySQL is running and the password in DB_PASSWORD is correct.")
727         return
728
729     # Each menu number is linked to the function that handles it
730     menu_actions = {
731         "1": add_record, "2": view_all_records, "3": search_menu,
732         "4": update_record, "5": delete_record, "6": overall_summary,
733         "7": category_summary, "8": monthly_summary, "9": highest_emission_category,
734         "10": export_to_csv, "11": export_summary_to_text,
735         "12": manage_emission_factors, "13": show_help, "14": undo_last_record,
736     }
737
738     try:
739         while True:
740             display_main_menu()
741             choice = input("\nEnter your choice: ").strip()
742             if choice == "0":
743                 print("Thank you for using Carbon Data Logger. Goodbye!")
744                 break
745             elif choice in menu_actions:
746                 try:
747                     menu_actions[choice]()
748                 except mysql.connector.Error as error:
749                     print("Database error:", error)
750                 # Sub-menus already pause after each action
751                 if choice not in ("3", "12"):
752                     pause_program()
753             else:
754                 print("Invalid choice! Please enter a number from 0 to 14.")
755     except (KeyboardInterrupt, EOFError):
756         print("\nProgram stopped by the user.")
757     finally:
758         # Always close the cursor and connection when the program ends
759         cursor.close()
760         connection.close()
761
762
763 main()
```